

UNIT 4 :Simplification of Boolean Algebra

Presented by

Er.Harendra Bikram Shah

(Lecturer)

OUTLINES

- ❖ K-Map of 2,3,4 variables
- ❖ Simplification and Realization using NAND and NOR gates
- ❖ Practical Design Steps

K-Map

Introduction to K-map:

- In previous chapters, we have simplified the Boolean functions using Boolean postulates and theorems.
- It is a time consuming process and we have to re-write the simplified expressions after each step.
- To overcome this difficulty, Karnaugh introduced a method for simplification of Boolean functions in an easy way.
- **A Karnaugh map (K-map) is a pictorial method used to minimize Boolean expressions without having to use Boolean algebra theorems and equation manipulations.**
- A K-map can be thought of as a special version of a truth table .
- It is a graphical method, which consists of 2^n cells for 'n' variables.
- In 1953, the American Physicist Maurice **Karnaugh** invented K-Map

K-Map

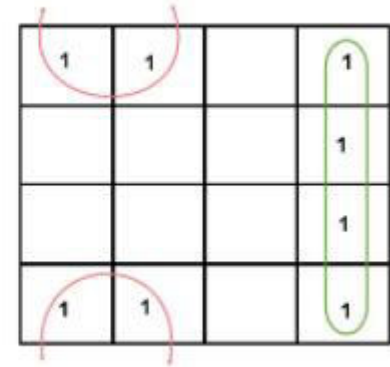
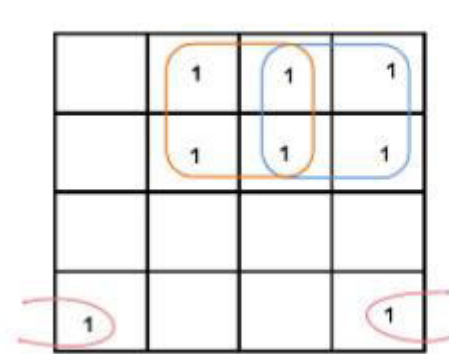
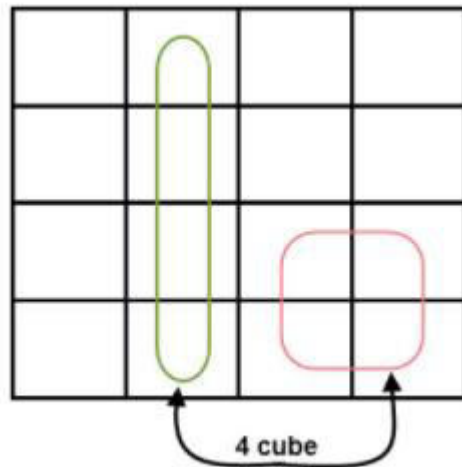
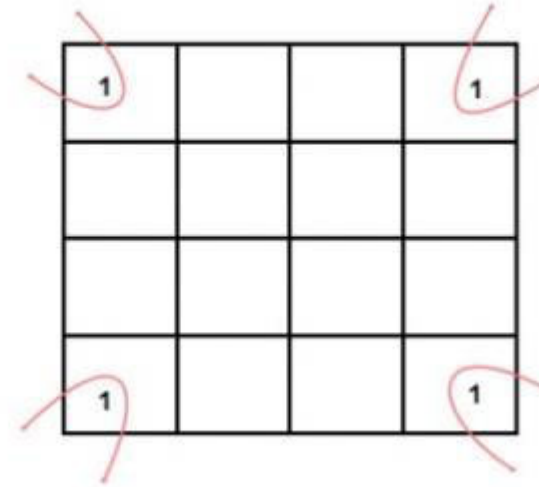
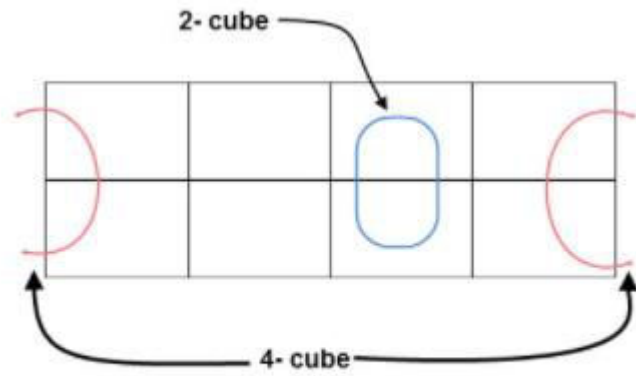
Significance/Application/Importance of K-map:

- Boolean algebra can be simplified systematically
- K-map reduces logic functions quickly and simply.
- K-maps are both faster and Easier for more number for variables like 4 or more numbers of variables.

Steps to solve expression using K-map-

1. Select K-map according to the number of variables.
2. Identify minterms or maxterms as given in problem.
3. For SOP put 1's in blocks of K-map respective to the minterms (0's elsewhere).
4. For POS put 0's in blocks of K-map respective to the maxterms (1's elsewhere).
5. Make rectangular groups containing total terms in power of two like 2, 4, 8 .. (except 1) and try to cover as many elements as you can in one group.
6. Don't care "x" should also be included while grouping to make a larger possible group.
7. From the groups made in step 5 find the product terms and sum them up for SOP form.

Pairing Ways in k-map



SOP FORM

K-Map of 2 variables

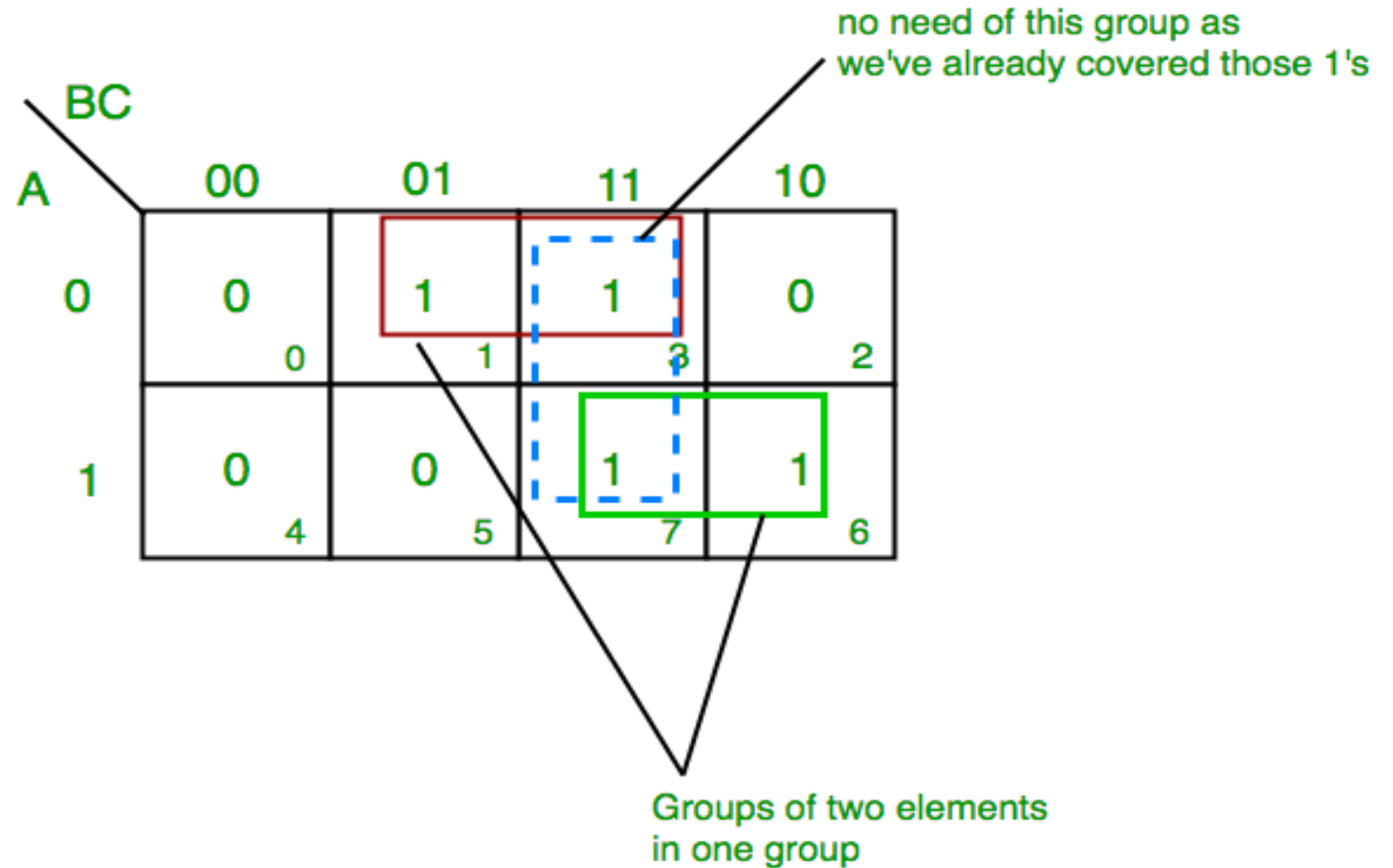
		B	
		0	1
A	0	$A'B'$ 0	$A'B$ 1
	1	AB' 2	AB 3

K-Map of 3 variables

SOP FORM

1.K-map of 3 variables-

$$Z = \sum A,B,C(1,3,6,7)$$



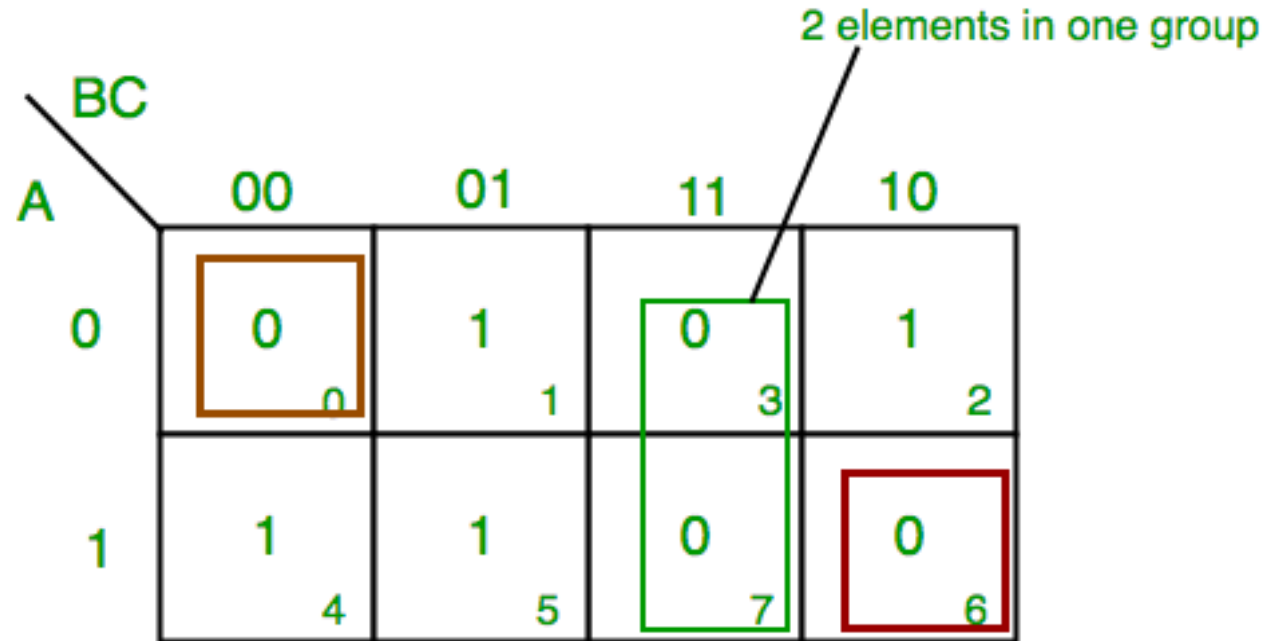
From **red** group we get product term—
 $A'C$

From **green** group we get product term
 AB

Summing these product terms we get- **Final expression**
 $(A'C + AB)$

SoP FORM

K-Map of 3 variables



SoP FORM

K-Map of 4 variables

		CD			
		$C'D'$	$C'D$	CD	CD'
A B	$A'B'$ 00	0	1	3	2
	$A'B$ 01	4	5	7	6
AB	11	12	13	15	14
	AB' 10	8	9	11	10

PoS FORM

K-Map of 4 variables

		C+D			
		C+D	C+D'	C'+D'	C'+D
A+B					
A+B 0+0		0	1	3	2
A+B' 0+1		4	5	7	6
A'B' 1+1		12	13	15	14
A'+B 1+0		8	9	11	10

5 Variable K-Map

The number of cells in 5 variable K-map is thirty-two, since the number of variables is 5. The following figure shows 5 variable K-Map.

- There is only one possibility of grouping 32 adjacent min terms.
- There are two possibilities of grouping 16 adjacent min terms. i.e.,
- grouping of min terms from m_0 to m_{15} and m_{16} to m_{31} .
- If $v=0$, then 5 variable K-map becomes 4 variable K-map.

		V=0			
		YZ			
WX		00	01	11	10
00		m_0	m_1	m_3	m_2
01		m_4	m_5	m_7	m_6
11		m_{12}	m_{13}	m_{15}	m_{14}
10		m_8	m_9	m_{11}	m_{10}

		V=1			
		YZ			
WX		00	01	11	10
00		m_{16}	m_{17}	m_{19}	m_{18}
01		m_{20}	m_{21}	m_{23}	m_{22}
11		m_{28}	m_{29}	m_{31}	m_{30}
10		m_{24}	m_{25}	m_{27}	m_{26}

In the above all K-maps, we used exclusively the min terms notation. Similarly, you can use exclusively the Max terms notation.

K-Map with Don't care

- Everything is same except the notation 'X' is used for don't care condition
- Don't care condition will remain same for both POS and SOP form